Ahmed Mohamed Ahmed

I am a First-Class Honors Computer Science graduate specializing in Artificial Intelligence. I have direct experience architecting and deploying end-to-end machine learning systems, with a technical focus on data engineering, computer vision, and Large Language Models. I enjoy applying a project-driven approach to solving complex problems.

Experience

Research Ready Intern - Google DeepMind (with University of Exeter) Jun 2025 – Aug 2025

Exeter, UK

Worked on "Scientific Paper Understanding with Multimodal LLMs and Knowledge Graphs" research project under the supervision of Dr. Hang Dong and Dr. Zhang Guioqang. Developed and evaluated methods to improve the quality and contextual accuracy of image captioning for figures within scientific papers. Worked with multimodal architectures to bridge the gap between visual (figures, graphs) and textual (paper content) information.
Designed prompt sets and a human evaluation rubric for scientific-figure captioning; piloted LLM-as-judge checks for factual consistency.
Built a lightweight eval harness for output correctness (batch runs, scoring, error taxonomy, reporting) to surface failure modes.

Junior Data Scientist Intern - Kelp Industries (Kelpi) Jan 2025 – June 2025

Bristol, UK

Given full ownership to architect, build, and deploy the company's primary data infrastructure from the ground up on Microsoft Azure. Led the complete technical delivery of the system, designing and implementing the PostgreSQL database, all custom ETL pipelines, and automated workflows for data ingestion, transformation, and validation. Standardized data formats and established validation processes, which reduced manual data cleanup by 80% and significantly accelerated experimental analysis timelines for the science teams. Worked directly and cross-functionally with chemists and research scientists, holding regular lab visits to gather requirements, iterate on data models, and roll out tailored solutions that mapped to their experimental processes. Acted as an ML advisor to the CTO and research teams, consulting on Bayesian optimization strategies for biomaterials experiments and helping translate domain research into an actionable technical roadmap.

Freelance ML Consultant - Various Clients 2024 – Present

Remote

For Patina Systems, a stealth-mode startup in New York, I designed and built a location-based image matching API using FastAPI to serve as a core component of their product.
For Pagerift, I fine-tuned a Large Language Model (Gemma 3) specifically for creative story writing, helped integrate the DSPy framework to improve model output quality, and provided general consulting on their backend development.

Machine Learning Engineer Intern - Noetic Health May 2024 – Aug 2024

Remote

Contributed to an AI-powered diagnostic platform for mental health by assisting in the integration of multi-agent systems and LLMs (OpenAI, Claude, Groq) to create dynamic and empathetic user assessments. Developed and maintained backend APIs using FastAPI and MongoDB, ensuring efficient and scalable communication between microservices. Implemented crucial safety and ethical guardrails to ensure the responsible and secure use of AI in a sensitive healthcare application. Utilized AWS services to support the deployment and optimize the scalability of the platform in a production environment.
Wrote prompt guidelines and implemented guardrails/safety gates for mental-health conversational flows.
Ran small A/B tests and collected human-in-the-loop safety ratings to calibrate thresholds and escalation rules.
Deployed FastAPI services with structured logging to capture evaluation traces for safety review.

Education

BSc Computer Science, First Class Honours - University of the West of England, Bristol, UK 2021 – 2025

Completed Key Modules: Advanced AI 3, Machine Learning, AI 2, Advanced Software Development, Advanced Algorithms, Autonomous Agents and Multi-Agent Systems, Distributed and Enterprise Software Development.

Technical Skills

Evaluation & Safety: LLM-as-judge, Eval harnesses (output correctness), Guardrails/safety gates

Languages: Python, SQL, C++

AI/ML: PyTorch, TensorFlow, Keras, Scikit-learn, FastAPI, Django, LLM Fine-Tuning, DSPy, Computer Vision (Object Detection, Image Retrieval), Multi-Agent Systems, Data Preprocessing and Optimization

Databases: PostgreSQL, MongoDB, NoSQL, SQL-based database design

Tools & Cloud: Azure, AWS, Git, GitHub, CircleCI, Jupyter Notebooks, Agile (Scrum)

Projects

BSc Dissertation: Automatic Detection of Diabetic Retinopathy on Edge Devices

Investigated and compared Vision Transformer (ViT) and CNN architectures for detecting diabetic retinopathy, with a specific focus on deployment to resource-constrained hardware.
Designed and implemented robust data-gathering and preprocessing pipelines for large-scale retinal image datasets to ensure valid training and evaluation splits.
Applied model optimization techniques, including quantization and pruning, to successfully reduce model size and run inference efficiently on embedded devices. Reported latency/size/accuracy trade-offs after quantization/pruning to inform edge deployment.
Authored a formal dissertation covering the end-to-end methodology, experimental analysis, results, and deployment lessons learned.

No Stylist App

Built machine learning backend for a fashion application.
Designed and implemented a product similarity image retrieval model using ResNet-50, experimenting with contrastive and triplet loss functions to achieve higher accuracy.
Developed a core feature using a YOLOv8n model for clothing detection in user-uploaded images.
Managed the optimization of large-scale image datasets and leveraged high-performance GPUs for model training and deployment.

Muse

Designed, built, and managed a natural-language + multimodal search turning a few image/text references into curated adjacent inspirations.
Implemented vibe-first retrieval with embeddings and lightweight descriptors; hybrid queries and diversity re-rank for fuzzy, intuitive exploration.
Delivered branching exploration and reproducible recipes (blend/regenerate, scope, min score, pin/ban) with de-duplication and session memory.
Optimized latency and UX with parallel candidate fetch and MMR; P95 under 2.5s on typical sets; clean controls and fast feedback.